Exp No. 3: Spectrum Analysis of Sub-1 GHz, Mid-band & mmWave by Plotting Propagation

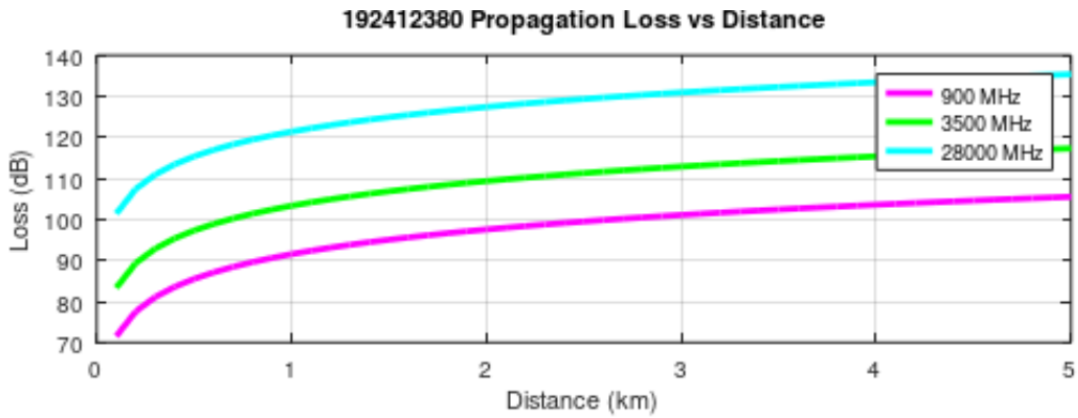
AIM: To analyze the propagation loss characteristics of different frequency bands (Sub-1 GHz, Midband, and mmWave) in wireless communication systems and compare their performance over varying distances using MATLAB.

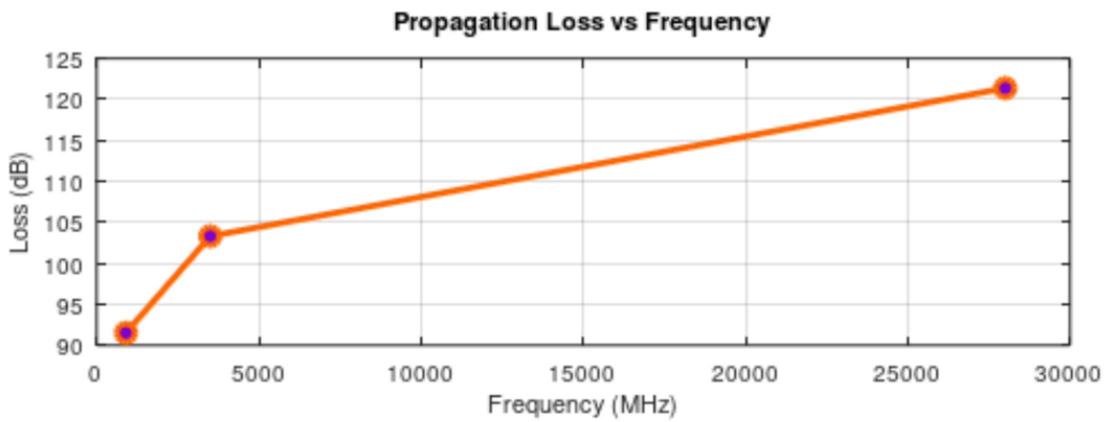
Objective 1: To compute the Propagation Loss (dB) for Sub-1 GHz, Mid-band, and mmWave frequency ranges using standard path loss models.

Program

```
clc;
clear;
close all;

% Distance (km)
d = 0.1:0.1:5;

% Frequencies (MHz)
f = [900 3500 28000];

% Free Space Path Loss (FSPL)
PL1 = 32.44 + 20*log10(f(1)) + 20*log10(d);
PL2 = 32.44 + 20*log10(f(2)) + 20*log10(d);
PL3 = 32.44 + 20*log10(f(3)) + 20*log10(d);

figure;

%------------- Graph 1 -------------%
subplot(2,1,1);

plot(d, PL1, 'm', 'LineWidth', 2);   % Magenta
hold on;
plot(d, PL2, 'g', 'LineWidth', 2);   % Green
plot(d, PL3, 'c', 'LineWidth', 2);   % Cyan

title('192412380 Propagation Loss vs Distance');
xlabel('Distance (km)');
ylabel('Loss (dB)');
grid on;

legend('900 MHz','3500 MHz','28000 MHz');

%------------- Graph 2 -------------%
subplot(2,1,2);
```

OUTPUT :


192412380 Propagation Loss vs Distance
Loss (dB)
Distance (km)
900 MHz
3500 MHz
28000 MHz
140
130
120
110
100
90
80
70
0
1
2
3
4
5

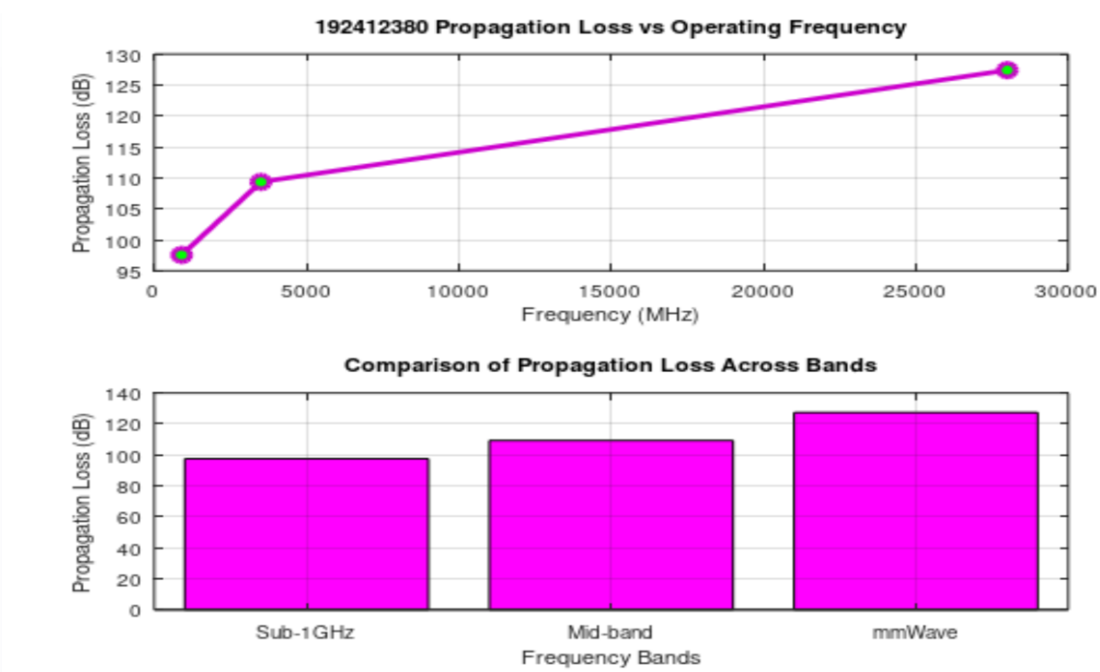

Propagation Loss vs Frequency
Loss (dB)
Frequency (MHz)
125
120
115
110
105
100
95
90
0
5000
10000
15000
20000
25000
30000

Objective 2: To evaluate the impact of different Operating Frequencies (GHz) on signal attenuation and coverage performance.

Program

```
clc;
clear;
close all;

% Fixed Distance (km)
d = 2;

% Operating Frequencies (MHz)
f = [900 3500 28000];

% Free Space Path Loss (FSPL)
PL = 32.44 + 20*log10(f) + 20*log10(d);

figure;

%------------ Graph 1 ------------%
subplot(2,1,1);

plot(f, PL, ...
    '-o', ...
    'Color', [0.8 0 0.8], ...      % Purple line
    'MarkerFaceColor', [0 1 0], ...% Green markers
    'LineWidth', 2);

title('192412380 Propagation Loss vs Operating Frequency');
xlabel('Frequency (MHz)');
ylabel('Propagation Loss (dB)');
grid on;

%------------ Graph 2 ------------%
subplot(2,1,2);

bar(PL, 'FaceColor', 'm');     % Magenta bars

```

OUTPUT :

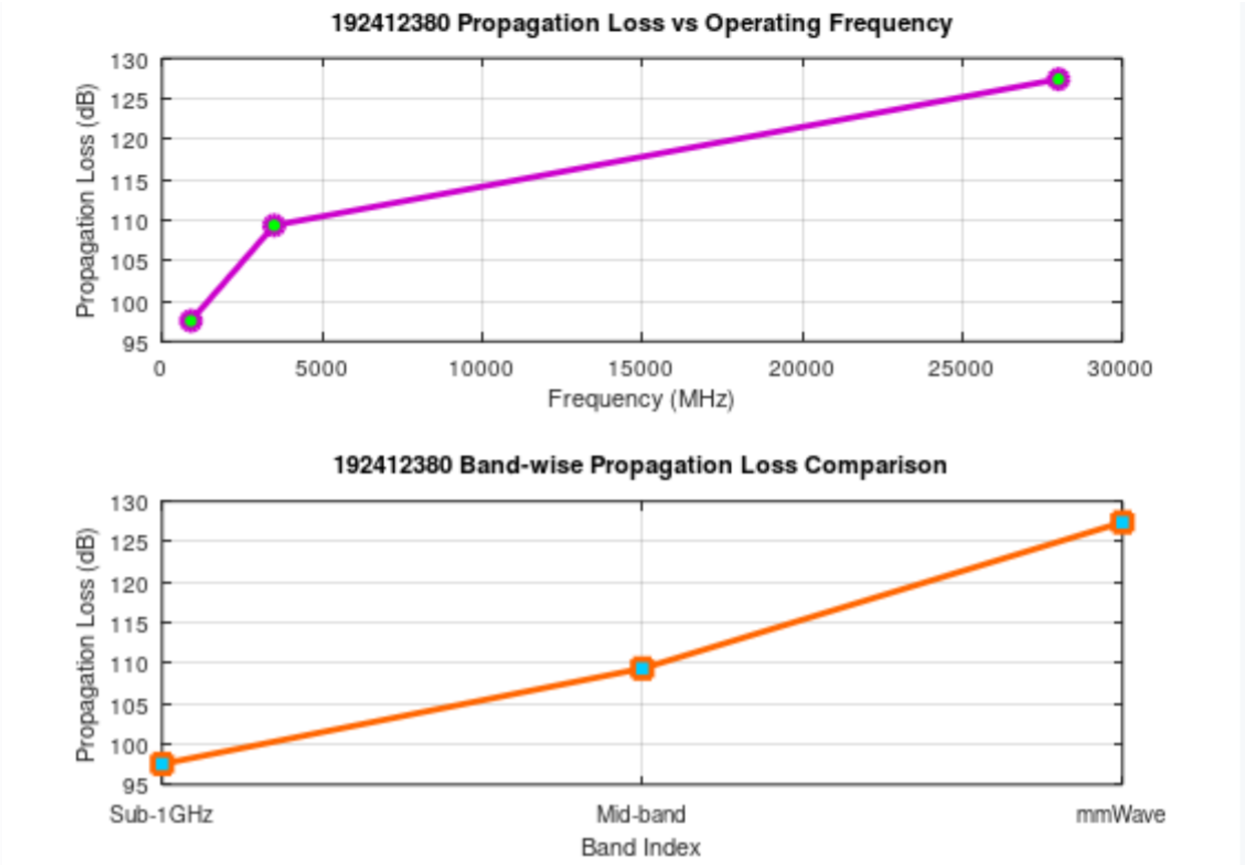


Objective 3: To analyze how **Communication Distance (km)** affects propagation loss across different spectrum bands.

program :

```
clc;
clear;
close all;

% Frequency bands (MHz)
f = [900 3500 28000];

% Distance (km)
d = 2;

% Free Space Path Loss (FSPL)
PL = 32.44 + 20*log10(f) + 20*log10(d);

figure;

%------------- Graph 1 -------------%
subplot(2,1,1);

plot(f, PL, ...
    '-o', ...
    'Color', [0.8 0 0.8], ...      % Purple line
    'MarkerFaceColor', [0 1 0], ...% Green markers
    'LineWidth', 2);

title('192412380 Propagation Loss vs Operating Frequency');
xlabel('Frequency (MHz)');
ylabel('Propagation Loss (dB)');
grid on;

%------------- Graph 2 -------------%
subplot(2,1,2);

plot(1:3, PL, ...
    '-s', ...
```

OUTPUT :



Result:

1.The propagation loss was successfully computed for Sub-1 GHz, Mid-band, and mmWave frequency bands using the Free Space Path Loss (FSPL) model.

2.It was observed that propagation loss increases with increase in operating frequency, with mmWave showing the highest attenuation and Sub-1 GHz showing the lowest loss.

3.The comparative analysis of spectrum bands clearly demonstrates that lower frequency bands are more suitable for long-range communication, while higher frequency bands require advanced techniques like beamforming for efficient performance.